Correlation coefficient

1

a 1

b Between -0.9 and -1

c Approximately 0

d Between -0.3 and -0.5

e Between 0.8 and 1

f Between 0.4 and 0.6

g Between -0.2 and 0.2

h Between -0.6 and -0.4

i Between -1 and -0.8

2

a As the heights of women increases, the probability of being turned down for a promotion decreases.

b As the population of a city increases, the number of speeding fines recorded increases.

c There is no evidence of a linear relationship between length of hair and length of fingernails.

d As the number of bylaws a council has about dog breeding increases, the number of dogs available for adoption at the local shelter tends to increase.

3

a

i $r = 0.47$

ii Weak, positive linear correlation

b

i $r = -0.92$

ii Strong, negative linear correlation

c

i $r = -0.84$

ii Strong, negative linear correlation

d

i $r = 0.85$

ii Strong, positive linear correlation

e

i $r = -0.59$

ii Moderate, negative linear correlation

f

i $r = 0.19$

ii Weak, positive linear correlation

g

i $r = -0.47$

ii Weak, negative linear correlation

h

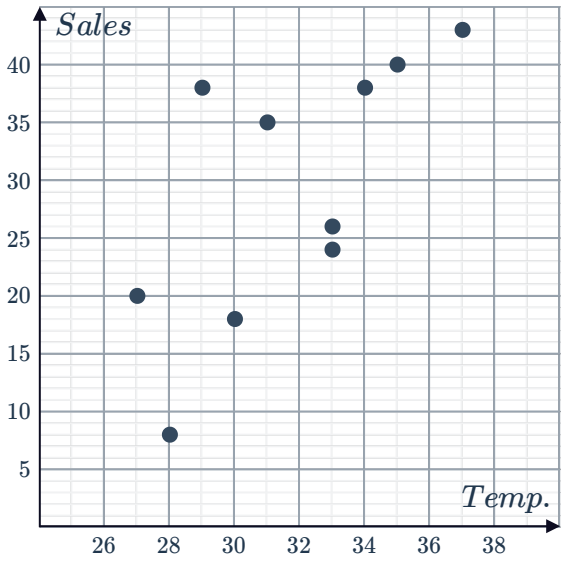
i $r = -0.84$

ii Strong, negative linear correlation

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4

a



b $r = 0.69$

c The sales tend to increase.

Correlation vs causation

5

a No

b Yes

6

a No

b Yes

c No

d No

7

a Greater than zero

b Yes, as temperature rises this would cause people to buy more fans.

8

a No. The high temperature may increase the number of people that go to the beach. This then increases the number of people that could drown.

b No. There is an indirect relationship between the variables.

9

a No

b Yes, there are no other contributing factors or reasonable arguments to be made for the strong positive association between the approximate number of pirates out at sea and the average world temperature.

c No

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10 Example answer:
An increase in cars is associated with an increase in strawberry sales.

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